Problem sheet 4 for Rigid Analytic Geometry Summer Term 2026

Let K be any field and $R \subseteq K$ a valuation ring of K. Recall that this means that R is a subring of K and for all $k \in K \setminus R$ we have $k^{-1} \in R$.

Problem 1 (1 point)**.** *For all $x, y \in K$, at least one of $x \in yR$ and $y \in xR$ holds.*

Let $\mathfrak{m}_R = \{0\} \cup \{k \in K^\times \mid k^{-1} \notin R\}$.

Problem 2 (3 points)**.** *Show the following:*

- *We have $\mathfrak{m}_R = R \setminus R^\times$.*
- *We have $R\mathfrak{m}_R \subseteq \mathfrak{m}_R$.*
- *If $x, y \in \mathfrak{m}_R$ then $x + y \in \mathfrak{m}_R$.*

It follows that $\mathfrak{m}_R$ is an ideal in R. Thus, the following (hopefully well-known) fact from algebra applies to R.

Problem 3 (3 points)**.** *For a ring R, the following conditions are equivalent:*

- *R has precisely one maximal ideal.*
- *$R \setminus R^\times$ is an ideal in R.*

In this case, $\mathfrak{m}_R = R \setminus R^\times$ is an ideal in R.

Recall that such rings are called local. We have thus shown that every valuation ring is a local ring.

Problem 4 (2 points)**.** *Show that R is integrally closed in K!*

Problem 5 (2 points)**.** *Show that the set of ideals of R is linearly ordered by $\subseteq$!*

Problem 6 (1 point)**.** *Show that every finitely generated ideal of R is principal!*

Problem 7 (2 points)**.** *An ideal $I \subseteq R$ is prime if and only if $1 \notin \mathfrak{p}$ and $\mathfrak{p} = \sqrt{I}$.*

Let A be any ring and $A \xrightarrow{v} \Gamma$ a valuation with support $\mathfrak{p} \in \mathrm{Spec}A$. It is obvious (Fact 1.3.3 from the lecture) that there is a unique valuation $A/\mathfrak{p} \to \Gamma$ with support $\{0\}$ such that v coincides with the composition $A \to A/\mathfrak{p} \to \Gamma$ and that this valuation of $A/\mathfrak{p}$ extends to the field of quotients $\mathfrak{k}(\mathfrak{p})$ of $\mathfrak{p}$ in a unique way. These valuations of $A/\mathfrak{p}$ and $\mathfrak{k}(\mathfrak{p})$ will also be denoted v. Let $R_v = \{x \in \mathfrak{k}(\mathfrak{p}) \mid v(x) \leq 1\}$ be the valuation ring of the valuation $\mathfrak{k}(\mathfrak{p}) \xrightarrow{v} \Gamma$. If V_A denotes the set

of pairs $(\mathfrak{p}, R)$ where $\mathfrak{p} \in \mathrm{Spec}A$ and R is a valuation ring of $\mathfrak{k}(\mathfrak{p})$, we have obtained a map

$$\mathrm{Spv}(A) \xrightarrow{(v/\sim) \to (\mathrm{supp}v, R_v)} V_A. \tag{1}$$

This is a bijection, with inverse sending $(R, \mathfrak{p})$ to the composition with $A \to \mathfrak{k}(\mathfrak{p})$ of the valuation of $\mathfrak{k}(\mathfrak{p})$ defined by R and Proposition 1.4.1 from the lecture. We denote this valuation of A by $v_{\mathfrak{p},R}$ and equip V_A with the topology for which (1) is a homeomorphism.

It is clear from the constructions that under this bijection that $\mathcal{R}(f_0|f_1, \ldots, f_m) \subseteq \mathrm{Spv}A$ corresponds to

$$\tilde{\mathcal{R}}(f_0|f_1, \ldots, f_m) = \big\{(\mathfrak{p}, R) \in V_A \;\big|\; f_0 \notin \mathfrak{p} \text{ and all } (f_i \bmod \mathfrak{p}) \in (f_0 \bmod \mathfrak{p}) \cdot R\big\}. \tag{2}$$

We can use this to give another description of generic limits of ultrafilters $\mathfrak{U}$ on $\mathrm{Spv}(R)$ and another proof of the fact that $\mathrm{Spv}(A)$ is a spectral space. Let

$$\mathfrak{p}_{\mathfrak{U}} = \Big\{a \in A \;\Big|\; \{(\mathfrak{p}, R) \in V_A \mid a \in \mathfrak{p}\} \in \mathfrak{U}\Big\}.$$

We have seen in Subsection 1.2 of the lecture that this is a prime ideal in A and is in fact the generic limit in $\mathrm{Spec}A$ of the direct image of $\mathfrak{U}$ under the map

$$V_A \xrightarrow{(\mathfrak{p}, R) \to \mathfrak{p}} \mathrm{Spec}A.$$

Let

$$R_{\mathfrak{U}} = \Big\{\frac{z \bmod \mathfrak{p}}{n \bmod \mathfrak{p}} \;\Big|\; z \in A \text{ and } n \in A \setminus \mathfrak{p}_{\mathfrak{U}} \text{ and } \big\{(\mathfrak{p}, R) \in V_R \;\big|\; v_{R,\mathfrak{p}}(z) \leq v_{R,\mathfrak{p}}(n)\big\} \in \mathfrak{U}\Big\}.$$

Problem 8 (4 points)**.** *Show that* $(\mathfrak{p}_{\mathfrak{U}}, R_{\mathfrak{U}})$ *is a generic limit of* $\mathfrak{U}$ *in* V_A*! Moreover, show that the open subsets of* V_A *of the form* (2) *are closed under generic limits of ultrafilters!*

In the following, the machinery of Grothendieck topologies will be applied to to $\mathfrak{O}_X$, the poset (with partial order $\subseteq$) of open subsets of a topological space X. If $U \in \mathfrak{O}_X$, a U-sieve then simply is a subset $\mathcal{S} \subseteq \mathfrak{O}_U$ such that $V \in \mathcal{S}$ implies $\mathfrak{O}_V \subseteq \mathcal{S}$. If $V \in \mathfrak{O}_U$, then the pull-back of $\mathcal{S}$ under the unique morphism $V \to U$ is

$$\mathcal{S}\,|_V := \mathcal{S} \cap \mathfrak{O}_V.$$

The axioms for a Grothendieck topology $\mathfrak{T}$ on $\mathfrak{O}_X$ become

GTTriv: For all $U \in \mathfrak{O}_X$, we have $\mathfrak{O}_U \in \mathfrak{T}_U$.

GTTrans: For all $U \in \mathfrak{O}_X$, all $V \in \mathfrak{O}_U$ and all $\mathcal{S} \in \mathfrak{T}_U$, we have $\mathcal{S}\,|_V \in \mathfrak{T}_V$.

GTLoc: For all $U \in \mathfrak{O}_X$ and all $\mathcal{S} \in \mathfrak{T}_U$, if $\mathcal{T}$ is an U-sieve such that $\mathcal{T}\,|_V \in \mathfrak{T}_V$ holds for all $V \in \mathfrak{S}$, then $\mathcal{T} \in \mathfrak{T}_U$.

Problem 9 (5 points)**.** *Let X be a topological space and $\mathfrak{B}$ a topology base for X such that for $\Omega, \Theta \in \mathfrak{B}$, the intersection $\Omega \cap \Theta$ is a finite union of elements of $\mathfrak{B}$. For $U \in \mathfrak{O}_X$, let $\mathfrak{B}_U = \mathfrak{B} \cap \mathfrak{O}_U$ and let $\mathfrak{T}_U$ be the set of all U-sieves $\mathcal{S}$ such that every $\Omega \in \mathfrak{B}_U$ is a union of finitely many elements of $\mathfrak{B}_\Omega \cap \mathcal{S}$. Show that $\mathfrak{T}$ is a Grothendieck topology on $\mathfrak{O}_X$ and that $U = \bigcup_{V \in \mathcal{S}} V$ holds for all $\mathcal{S} \in \mathfrak{T}_U$!*

Note that the last assertions simply asserts that all covering sieves are coverings in the ordinary topological sense. We call this Grothendieck topology on $\mathfrak{O}_X$ *the Grothendieck topology enforcing the quasicompactness of the elements of* $\mathfrak{B}$.

Let K be a field with an absolute value $|\cdot|$ satisfying the conditions from sheet 1 with the possible exception of completeness. Let $\mathbb{K}$ denote the completion of K.

Problem 10 (5 points)**.** *Show that every finite Galois extension $\mathbb{L}$ of $\mathbb{K}$ has a dense subfield L which is a finite Galois extension of K!*

In particular, $\mathbb{K}$ is separably closed if K is. Together with the results from the previous sheet this shows that $\mathbb{K}$ is algebraically closed if K is algebraically closed. In particular, the completion $\mathbb{C}_p$ of the algebraic closure of $\mathbb{Q}_p$ with respect to the unique extension of $|\cdot|_{\mathbb{Q}_p}$ is algebraically closed.

Problem 11 (1 point)**.** *If X is any topological space and A the ring of continuous functions $X \xrightarrow{f} K$, then the map $X \to \mathrm{Spv}(A)$ sending x to the equivalence class of the valuation*

$$v_x(f) = |f(x)| \in [0, \infty)_{\mathbb{R}}$$

of A is continuous!

Let $\mathbb{T}_n = \mathbb{T}_{n,K}$. We denote the set of maximal ideals $\mathfrak{m} \subseteq \mathbb{T}_n$ by $\mathrm{Sp}\mathbb{T}_n$. On the previous sheet the crucial fact that $\mathfrak{k}(\mathfrak{m})$ is a finite extension of K for all $\mathfrak{m} \in \mathrm{Sp}\mathbb{T}_n$ was mentioned (it follows from Proposition 1 on that sheet) but not shown. For the time being we are going to simply use this fact, which will later on be shown in the lectures. It follows from this fact and the results on the third sheet that the absolute value on K has a unique extension to an absolute value on $\mathfrak{k}(\mathfrak{m})$ which will be denoted $|\cdot|_{\mathfrak{m}}$. For $f \in \mathbb{T}_n$, let $f(\mathfrak{m}) := f \bmod \mathfrak{m}$. We

get a map

$$\begin{aligned}(3)\qquad &\mathrm{Sp}\mathbb{T}_n \to \mathrm{Spv}\mathbb{T}_n\\ &\mathfrak{m} \to (\text{equivalence class of the valuation } f \to |f(\mathfrak{m})|_{\mathfrak{m}})\end{aligned}$$

and equip $\mathrm{Sp}\mathbb{T}_n$ with the coarsest topology for which it is continuous. A topology base of $\mathrm{Sp}\mathbb{T}_n$ is therefore given by the set of subsets

$$\begin{aligned}(4)\qquad \mathcal{R}_{\mathrm{Sp}\mathbb{T}_n}(f_0|f_1,\ldots,f_m) = \big\{x \in \mathrm{Sp}\mathbb{T}_n \;\big|\; f(x) \neq 0\\ \text{and } |f_i(x)|_x \leq |f_0(x)|_x \text{ for all } i\big\}\end{aligned}$$

with $(f_i)_{i=0}^m$ running over all of $\mathbb{T}_n^{m+1}$. A subset of $\mathrm{Sp}\mathbb{T}_n$ *rational* if it has the form (4) where the f_i are such that the ideal $I = \langle f_0,\ldots,f_m\rangle_{\mathbb{T}_n}$ equals $\mathbb{T}_m$. Since $\mathbb{T}_m$ has no proper open ideals (Problem 5 on sheet 1), an equivalent condition is that the ideal I is open. If this holds, (4) simplifies to

$$(5)\quad \mathcal{R}_{\mathrm{Sp}\mathbb{T}_n}(f_0|f_1,\ldots,f_m) = \big\{x \in \mathrm{Sp}\mathbb{T}_n \;\big|\; |f_i(x)|_x \leq |f_0(x)|_x \text{ for all } i\big\}.$$

Let $\mathfrak{Rat}_{\mathbb{T}_n}$ be the set of rational open subsets of $\mathrm{Sp}\mathbb{T}_n$.

Problem 12 (2 Points)**.** *Show that $\mathfrak{Rat}_{\mathbb{T}_n}$ is a topology base for $\mathbb{T}_n$.*

The main point here is that K has arbitrarily small non-zero elements which can be added to the list of f_i, meaning that arbitrary subsets of the form (4) can be expressed as a countably infinite union of elements of $\mathfrak{Rat}_{\mathbb{T}_n}$.

Problem 13 (2 points)**.** *Show that $\mathfrak{Rat}_{\mathbb{T}_n}$ is closed under finite intersections in $\mathrm{Sp}\mathbb{T}_n$!*

We equip $X = \mathrm{Sp}\mathbb{T}_n$ with the Grothendieck topology on $\mathfrak{O}_X$ enforcing the quasicompactness of the elements of $\mathfrak{Rat}_{\mathbb{T}_n}$, obtained by applying Problem 9 to the topology base $\mathfrak{Rat}_{\mathbb{T}_n}$ of X. This is the basic Grothendieck topology for the classical approach to rigid analytic geometry. Thus, all subsets of the form 4 with arbitrary $(f_i)_{i=0}^m \in \mathbb{T}_n^{m+1}$ are open, but the Grothendieck topology only enforces the quasicompactness of the elements of $\mathfrak{Rat}_{\mathbb{T}_n}$.

If K is algebraically closed (eg, $K = \mathbb{C}_p$) then the map

$$\begin{aligned}(6)\qquad &K^{\circ n} \to \mathrm{Sp}\mathbb{T}_n\\ &x \to \mathfrak{m}_x = \big\{f \in \mathbb{T}_n \;\big|\; f(x) = 0\big\}\end{aligned}$$

is bijective, by Proposition 1 on sheet 4.

Problem 14 (2 points)**.** *Under the identification* (6)*, the topology on $\mathrm{Sp}\mathbb{T}_n$ is identified with the topology defined by the metric defined by the norm*

$$\|(x_i)_{i=1}^n\,|K^n\| = \max_{1\leq i\leq n} |x_i|\,.$$

Of the 35 points from this sheet, 15 are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday June 1. Good luck and nice holidays!